

AI 4403 : Binomial Models For Option Pricing

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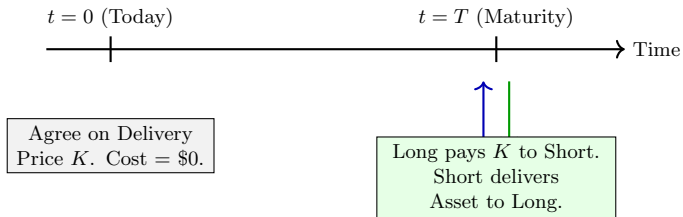
- 1 Forwards
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Forwards

What is a Forward Contract?

A legally binding agreement between two parties to buy or sell an asset at a specified future time (T) for a price agreed upon **today** (K).

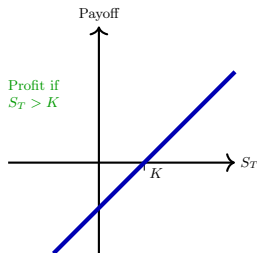
- ▶ **Long Position:** Agrees to *buy* the asset. (e.g., The Baker)
- ▶ **Short Position:** Agrees to *sell* the asset. (e.g., The Farmer)
- ▶ **Crucial Feature:** *No money changes hands today ($t = 0$).* The contract value at initiation is zero.



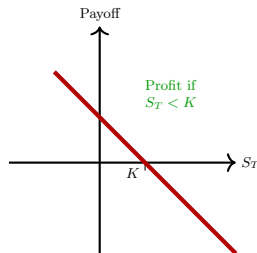
Payoff Profiles: The Linear Nature of Forwards

- ▶ At maturity T , the actual market price of the asset is S_T .
- ▶ Because they locked in price K , their profit/loss depends entirely on the difference between S_T and K .

$$\text{Long Payoff} = S_T - K$$



$$\text{Short Payoff} = K - S_T$$



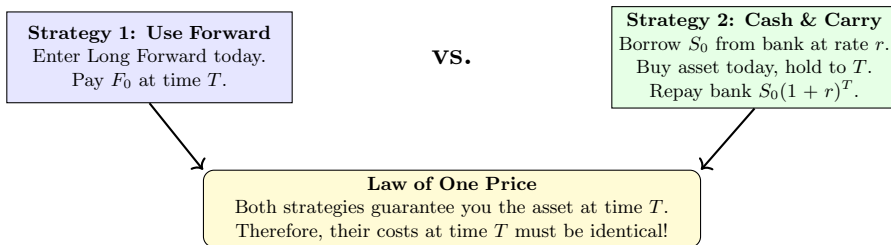
Key Observation

This payoff is perfectly **linear** and symmetric. One party's gain is exactly the other party's loss.

Pricing a Forward: The Arbitrage Argument

- ▶ How do we determine the fair delivery price (F_0) today so that the initial value of the contract is exactly \$0?
- ▶ **Do we need to guess where the stock price S_T is going? No!**
- ▶ We use **Static Replication** (Cash-and-Carry Arbitrage).

Suppose you want to own the asset at time T . You have two choices today ($t = 0$):



The Fundamental Forward Equation

Equating the costs of Strategy 1 and Strategy 2 at maturity T :

$$F_0 = S_0(1 + r)^T$$

- ▶ F_0 = Fair Forward Price
- ▶ S_0 = Current Spot Price of the underlying asset
- ▶ r = Risk-free interest rate
- ▶ T = Time to maturity

The Crucial Paradigm Shift

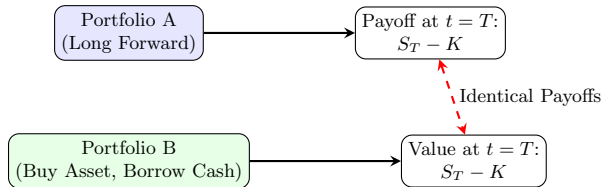
Notice what is **missing** from this formula. We priced a derivative without ever calculating the expected future price of the stock, probabilities, or risk aversion!

Because the payoff is *linear*, we can price it perfectly using just the **spot price** and the **bond (borrowing) rate**.

Pricing via Static Replication

To find the fair forward price F_0 , we use the **Law of One Price**. We create a portfolio today that perfectly matches the forward contract's payoff at maturity T .

- **Portfolio A (Long Forward):** Commit to buy asset at K . Cost at $t = 0$ is 0.
- **Portfolio B (Replicating Portfolio):** Buy the asset at S_0 and borrow the present value of the strike price Ke^{-rT} .



Conclusion: Since the payoffs match, the costs must match.

$$0 = S_0 - Ke^{-rT} \implies \text{Fair Price } \mathbf{F_0 = S_0 e^{rT}}$$

Note: This is a **static replication**. Once the portfolio is set up at $t = 0$, no dynamic rebalancing is required to match the final payoff.

Numerical Example: Setup

Let's define the market parameters for our underlying asset:

Market Parameters
Spot Price (S_0): \$100
Risk-Free Rate (r): 5% (continuous)
Time to Maturity (T): 1 year

First, we calculate the **Fair Forward Price** (F_0) using our replication formula:

$$F_0 = S_0 e^{rT} = 100 \times e^{0.05 \times 1} = 100 \times 1.05127 = \$105.13$$

The Rule:

- ▶ If $F_{\text{mkt}} > \$105.13$, the forward is too expensive $\rightarrow$ *Sell the forward.*
- ▶ If $F_{\text{mkt}} < \$105.13$, the forward is too cheap $\rightarrow$ *Buy the forward.*

Cash-and-Carry Arbitrage ($F_{\text{mkt}} > F_0$)

Scenario: The market is pricing the 1-year forward contract at $F_{\text{mkt}} = \$110$.
(Overpriced relative to fair value of \$105.13)

Action	Cash Flow Today ($t = 0$)	Cash Flow at $t = 1$
1. Borrow \$100 at 5%	+\$100.00	-\$105.13 (Repay loan)
2. Buy asset at spot	-\$100.00	Asset held
3. Short forward at \$110	\$0.00	+\$110.00 (Deliver asset)
Net Position	\$0.00	+\$4.87 (Risk-Free Profit)

Result: By shorting the overpriced derivative and going long the physical asset, you lock in a guaranteed profit of **\$4.87** without using your own capital.

Reverse Cash-and-Carry ($F_{\text{mkt}} < F_0$)

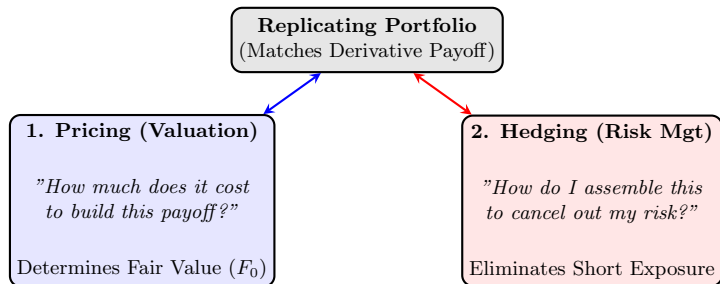
Scenario: The market is pricing the 1-year forward contract at $F_{\text{mkt}} = \$100$.
(Underpriced relative to fair value of \$105.13)

Action	Cash Flow Today ($t = 0$)	Cash Flow at $t = 1$
1. Short asset at spot	+\$100.00	Owe asset
2. Invest \$100 at 5%	-\$100.00	+\$105.13 (Receive cash)
3. Long forward at \$100	\$0.00	-\$100.00 (Receive asset)
Net Position	\$0.00	+\$5.13 (Risk-Free Profit)

Result: By buying the underpriced derivative and shorting the physical asset, you return the asset using the forward delivery and keep the **\$5.13** interest spread.

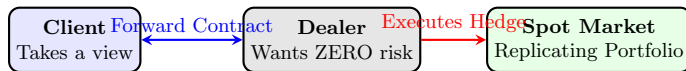
The Dual Purpose of Replication

Replication is the mathematical bridge connecting two fundamental concepts in quantitative finance. They are two sides of the same coin.



- **For the Theorist:** It proves the derivative's exact price using the Law of One Price.
- **For the Practitioner:** It provides a real-world trading recipe to eliminate risk when selling the derivative.

Financial institutions act as market makers. They take the opposite side of client trades but do not want exposure to market movements. They use replication to **neutralize** their risk.



- ▶ If Client goes **Long Forward**, Dealer is stuck with **Short Forward** risk.
- ▶ **The Fix:** Dealer immediately enters the **Long Replicating Portfolio** in the spot market to cancel the exposure.

Hedging a Short Forward Exposure

The Situation: The dealer sold a forward to a client (Dealer is **Short**).

The Risk: If S_T skyrockets, the dealer loses infinitely ($K - S_T$).

The Hedge: The dealer creates a Long Replicating Portfolio.

Dealer's Actions	Cash Flow at $t = 0$	Value at $t = T$
1. Client Trade (Short Fwd)	\$0	$K - S_T$
Hedge Execution:		
2. Borrow Ke^{-rT} cash	$+Ke^{-rT}$	$-K$ (Repay loan)
3. Buy asset in spot market	$-S_0$	$+S_T$ (Asset value)
Net Position	$Ke^{-rT} - S_0$	0 (Perfect Hedge)

Result: At maturity, the dealer hands the physical asset (which they already bought) to the client, collects K , and uses K to repay the loan. **Market risk is entirely eliminated.**

Hedging a Long Forward Exposure

The Situation: The dealer bought a forward from a client (Dealer is **Long**).

The Risk: If S_T drops to zero, the dealer faces a maximum loss of $-K$.

The Hedge: The dealer creates a Short Replicating Portfolio.

Dealer's Actions	Cash Flow at $t = 0$	Value at $t = T$
1. Client Trade (Long Fwd)	\$0	$S_T - K$
Hedge Execution:		
2. Short-sell the asset	$+S_0$	$-S_T$ (Must buy back)
3. Invest cash at rate r	$-S_0$	$+S_0e^{rT}$ (Receive cash)
Net Position	\$0	$S_0e^{rT} - K$

Result: At maturity, the dealer receives S_0e^{rT} from the investment. They use K to buy the asset from the client via the forward contract, and use that asset to close out their short position. **Any downside risk is neutralized.**

Options

- ▶ With Forwards and Futures, both parties are **obligated** to transact at maturity T .
- ▶ What if you only want to transact if the market moves in your favor?

What is an Option?

A contract that gives the buyer the **right, but not the obligation**, to buy or sell an underlying asset at a specified strike price (K) on or before a specified date (T).

Call Option

- ▶ The right to **BUY**.
- ▶ You want the asset price (S_T) to go **UP**.

Put Option

- ▶ The right to **SELL**.
- ▶ You want the asset price (S_T) to go **DOWN**.

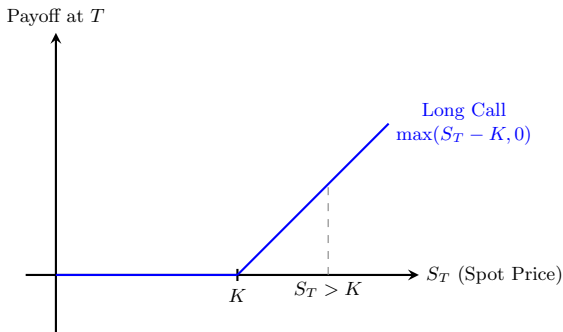
The Catch: The Premium

Because the buyer has a choice (and the seller has all the risk), entering an option contract is **not free**. The buyer must pay a price upfront at $t = 0$, called the **Option Premium**. Our goal is to figure out how to calculate this premium!

Call Option: Payoff Profile

A **Call Option** gives the holder the *right, but not the obligation*, to buy the underlying asset at a specified strike price (K) at maturity (T).

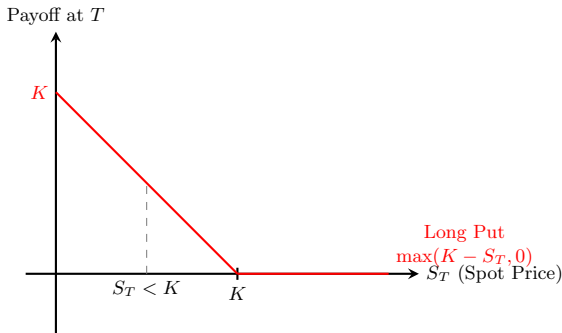
- **Long Call Payoff:** $\max(S_T - K, 0)$
- **Logic:** If $S_T > K$, the option is exercised (buy at K , worth S_T). If $S_T \leq K$, the option expires worthless (loss is limited to the initial premium paid, which is not shown in the raw terminal payoff).



Put Option: Payoff Profile

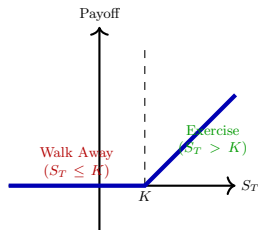
A **Put Option** gives the holder the *right, but not the obligation*, to sell the underlying asset at a specified strike price (K) at maturity (T).

- **Long Put Payoff:** $\max(K - S_T, 0)$
- **Logic:** If $S_T < K$, the option is exercised (sell at K , even though market is lower). If $S_T \geq K$, the option expires worthless.

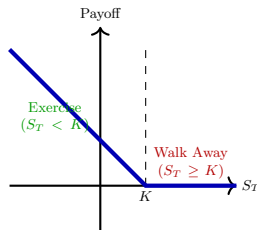


- At maturity T , the option buyer will only exercise the contract if it is profitable. Otherwise, they walk away (payoff is zero).

$$\text{Long Call Payoff} = \max(S_T - K, 0)$$



$$\text{Long Put Payoff} = \max(K - S_T, 0)$$

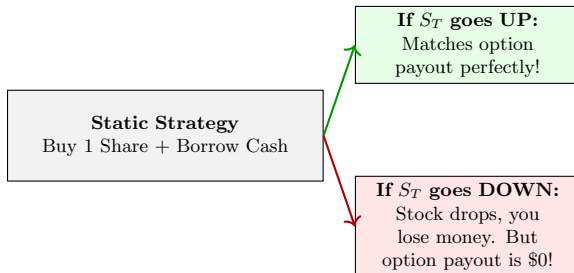


The Asymmetry

Unlike Forwards, the payoff is **kinked** at the strike price K . The downside is strictly limited to zero, while the upside can be massive.

The Breakdown of Static Replication

- **Recall:** We priced Forwards easily using a “Cash and Carry” strategy (borrow money, buy 1 share of stock today, hold until T).
- **Can we do this for a Call Option?**



The Pricing Dilemma

You **cannot** statically mix a straight line (stock) and a flat line (bond) to perfectly match a kinked curve (option) across all possible futures. The static arbitrage argument fails!

A Call Option's payoff is conditional: $\max(S_T - K, 0)$. If we had a magic oracle at $t = 0$ telling us the future, static replication would be easy.

World A: Oracle says it finishes ITM

- **Condition:** $S_T > K$
- **Payoff:** Exactly $S_T - K$
- **Static Replication:** Same as a Forward contract! Buy 1 unit of asset, borrow Ke^{-rT} .
- **Cost today:** $S_0 - Ke^{-rT} > 0$

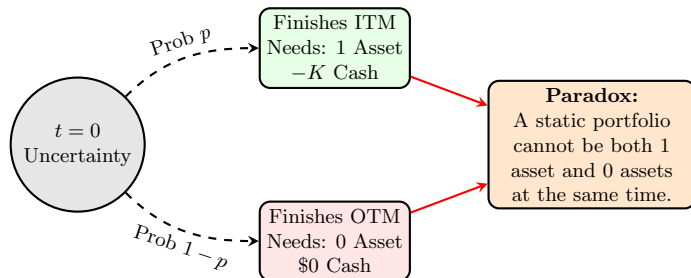
World B: Oracle says it finishes OTM

- **Condition:** $S_T \leq K$
- **Payoff:** Exactly 0
- **Static Replication:** Do nothing. Hold 0 units of the asset, borrow \$0.
- **Cost today:** \$0

The Catch: In these hypothetical worlds, the payoff is linear ($S_T - K$ or 0). Linear payoffs are perfectly replicated statically.

The Reality: Why Static Replication Fails

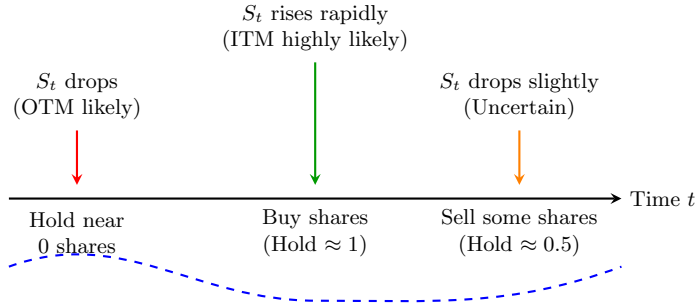
In reality, no oracle exists. At $t = 0$, we face **uncertainty** and must prepare for both possibilities simultaneously.



- ▶ If you build the ITM portfolio statically, and the market crashes, you lose S_0 (Option buyer only loses premium).
- ▶ If you build the OTM portfolio statically (do nothing), and the market skyrockets, you are exposed to infinite risk.

The Bridge to Dynamic Replication

Because a single, fixed portfolio cannot bridge this uncertainty, we must use a **Dynamic Replicating Portfolio** that adapts as the probabilities shift over time.



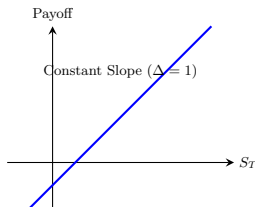
- This continuously changing ratio of "shares held" is known as the option's **Delta** (Δ).
- By constantly rebalancing, the portfolio "morphs" smoothly between the World A (ITM) and World B (OTM) setups.

Types of Replication: Static vs. Dynamic

The structure of the derivative's payoff dictates how it must be replicated.

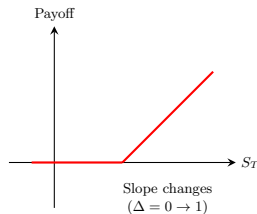
1. Static Replication

- **Action:** Set up portfolio at $t = 0$, leave untouched until T .
- **Applies to:** Linear payoffs (e.g., Forward Contracts).
- **Why?** The sensitivity (Δ) to the underlying asset is constant.



2. Dynamic Replication

- **Action:** Continuously rebalance the portfolio as spot price changes.
- **Applies to:** Non-linear/Kinked payoffs (e.g., Options).
- **Why?** The sensitivity (Δ) changes abruptly based on spot price.

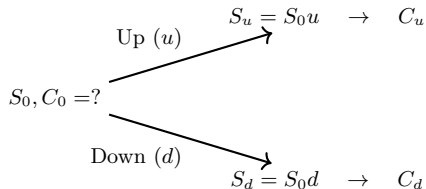


Option Pricing: Binomial Model

The One-Period Binomial Framework

Assume a stock today trading at S_0 . Over time step Δt , it can only move to two states:

$$\begin{cases} S_u = S_0 \cdot u & \text{with payoff } C_u = \max(S_u - K, 0) \\ S_d = S_0 \cdot d & \text{with payoff } C_d = \max(S_d - K, 0) \end{cases}$$



The Replicating Portfolio

Construct a portfolio of Δ shares of stock and B dollars invested in risk-free bonds yielding rate r .

Solving for Delta (Δ) and Bond (B)

To eliminate all risk, match the portfolio payoff to the option payoff in **both** states:

$$\begin{cases} \Delta S_u + Be^{r\Delta t} = C_u & (\text{Up State}) \\ \Delta S_d + Be^{r\Delta t} = C_d & (\text{Down State}) \end{cases}$$

Subtracting the second equation from the first yields the fundamental option hedge ratio:

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$

- ▶ Δ represents the **number of stock shares** needed to hedge one option.
- ▶ Since $S_u > S_d$ and $C_u \geq C_d$, we always have $0 \leq \Delta \leq 1$ for a European Call.
- ▶ By No-Arbitrage, the option price today must equal the cost of this portfolio:

$$C_0 = \Delta S_0 + B$$

The Core Breakthrough: Risk-Neutral Valuation

Substituting $B = e^{-r\Delta t}(C_u - \Delta S_u)$ back into $C_0 = \Delta S_0 + B$ yields:

$$C_0 = e^{-r\Delta t} \left[\left(\frac{e^{r\Delta t} - d}{u - d} \right) C_u + \left(\frac{u - e^{r\Delta t}}{u - d} \right) C_d \right]$$

Define the synthetic variable p^* :

$$p^* = \frac{e^{r\Delta t} - d}{u - d}$$

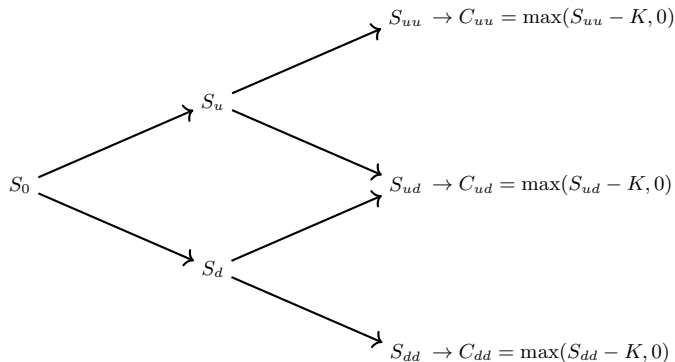
The Risk-Neutral Pricing Formula

$$C_0 = e^{-r\Delta t} [p^* C_u + (1 - p^*) C_d]$$

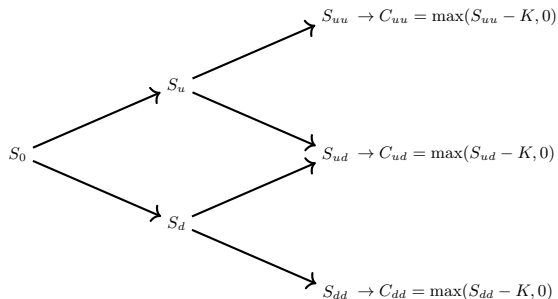
- ▶ p^* is **not** the physical probability of the stock moving up.
- ▶ p^* is the **risk-neutral probability**—the mathematical probability under which the expected return of the stock equals the risk-free rate r .

Scaling Up: The 2-Period Binomial Tree

To model realistic time horizons, we concatenate single periods into a multi-step tree ($t_0 \rightarrow t_1 \rightarrow t_2$).



Scaling Up: The 2-Period Binomial Tree



Backward Induction Protocol

1. **Terminal Nodes (t_2):** Evaluate intrinsic option payoffs directly at expiry.
2. **Intermediate Nodes (t_1):** Discount expected risk-neutral values:

$$C_u = e^{-r\Delta t}[p^*C_{uu} + (1 - p^*)C_{ud}], \quad C_d = e^{-r\Delta t}[p^*C_{ud} + (1 - p^*)C_{dd}]$$

3. **Root Node (t_0):** Step backward once more to obtain present option price C_0 .

Static replication fails because the required hedge ratio (Δ) changes at every node across the tree!

Hedge Ratio at Node $t = 0$

$$\Delta_0 = \frac{C_u - C_d}{S_u - S_d}$$

Hold Δ_0 shares of stock to stay hedged over $t_0 \rightarrow t_1$.

Hedge Ratio at Node $t = 1$ (Up State)

$$\Delta_u = \frac{C_{uu} - C_{ud}}{S_{uu} - S_{ud}}$$

Because $\Delta_u \neq \Delta_0$, the portfolio is no longer hedged!

Why Option Replication Must Be Dynamic

- ▶ As S_t moves, the likelihood of finishing in-the-money changes $\rightarrow \Delta$ shifts continuously.
- ▶ **Dynamic Rebalancing:** The market maker must actively buy or sell shares at intermediate nodes to maintain a zero-risk portfolio.

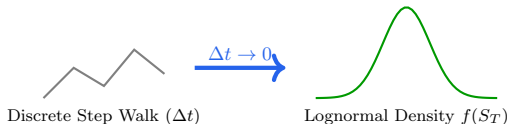
What happens when we shrink the time step $\Delta t \rightarrow 0$ and let the number of steps $N \rightarrow \infty$?

Discrete Binomial ($N = 4$)

Discrete coin flips yield coarse, step-wise payoff probabilities across a small set of outcome nodes.

Continuous Limit ($N \rightarrow \infty$)

By the Central Limit Theorem, the sum of infinitely many tiny binomial steps converges smoothly to a **Lognormal Distribution**.



Next lecture, we leave discrete coin flips behind and enter continuous time.

What's Coming Next?

1. **Continuous Random Walks:** Replacing u and d with Standard Brownian Motion (dW_t).
2. **Stochastic Calculus:** Standard differential calculus ($df = f'dx$) fails on non-smooth paths $\rightarrow$ introducing **Itô's Lemma**.
3. **Geometric Brownian Motion (GBM):** Modeling stock price dynamics continuous-time:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

4. **The Black-Scholes Partial Differential Equation (PDE):** Continuous dynamic delta-hedging yields the legendary formula:

$$C_0 = S_0 N(d_1) - K e^{-rT} N(d_2)$$